

W3&4 – Combinations of Functions and Inverse Functions

MHF4U

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1) Let $f(x) = 3x - 5$ and $g(x) = 2x + 3$.

a) Write the equation for $h(x) = f(x) + g(x)$ and determine the value of $h(2)$.

$$h(x) = (3x - 5) + (2x + 3)$$

$$h(2) = 5(2) - 2$$

$$h(x) = 3x - 5 + 2x + 3$$

$$h(2) = 8$$

$$h(x) = 5x - 2$$

b) Write the equation for $k(x) = f(x) - g(x)$ and determine the value of $k(2)$.

$$k(x) = (3x - 5) - (2x + 3)$$

$$k(2) = 2 - 8$$

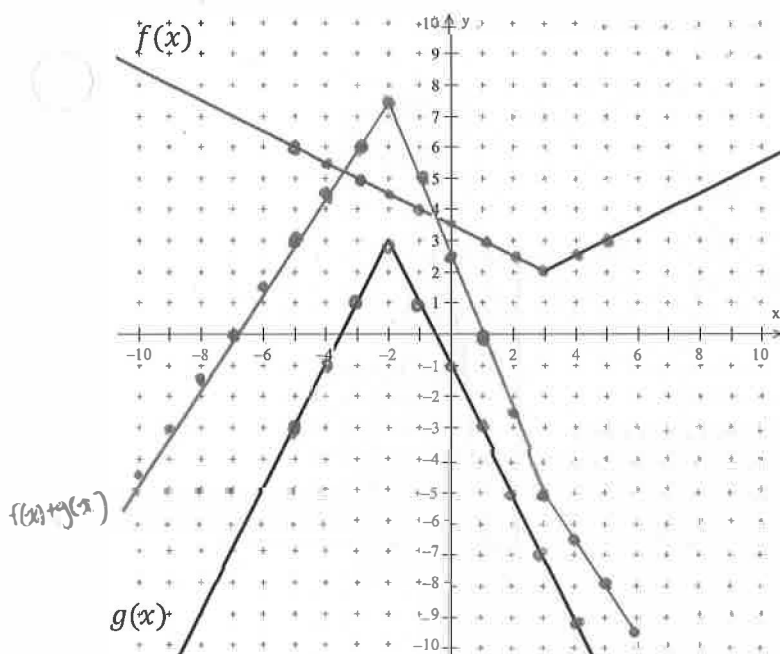
$$k(x) = 3x - 5 - 2x - 3$$

$$k(2) = -6$$

$$k(x) = x - 8$$

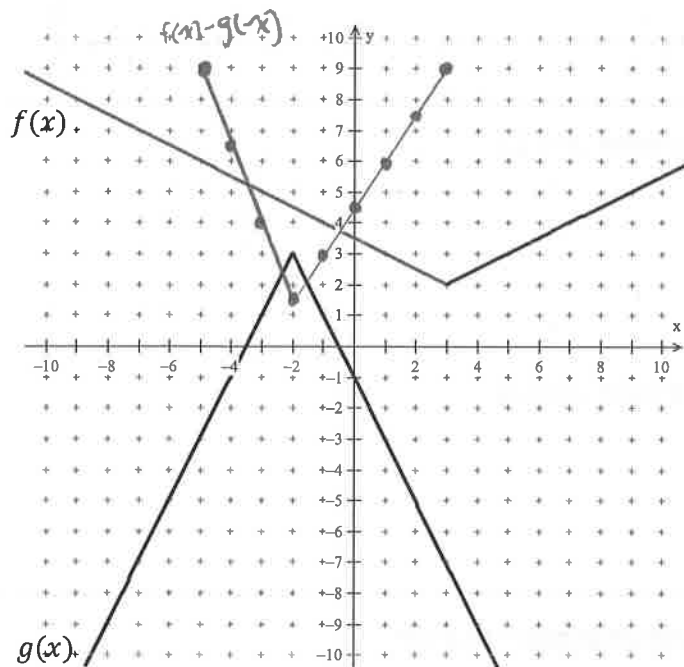
2) Use the functions $f(x)$ and $g(x)$ as shown. Apply the superposition principle to graph

a) $y = f(x) + g(x)$



x	$f(x)$	$g(x)$	$f(x) + g(x)$	$f(x) - g(x)$
-5	6	-3	3	9
-4	5.5	-1	4.5	6.5
-3	5	1	6	4
-2	4.5	3	7.5	1.5
-1	4	1	5	3
0	3.5	-1	2.5	4.5
1	3	-3	0	6
2	2.5	-5	-2.5	7.5
3	2	-7	-5	9
4	2.5	-9	-6.5	11.5
5	3	-11	-8	14

b) $y = f(x) - g(x)$.



3) Let $f(x) = x - 2$ and $g(x) = x^2 + 3x - 3$. Determine an algebraic and graphical model for $h(x) = f(x) + g(x)$.

$$h(x) = (x - 2) + (x^2 + 3x - 3)$$

$$h(x) = x - 2 + x^2 + 3x - 3$$

$$h(x) = x^2 + 4x - 5$$

$$h(x) = (x + 5)(x - 1)$$

x-int: $x = -5$ and $x = 1$

x-vertex at $\frac{-5+1}{2} = -2$

$$h(-2) = (-2)^2 + 4(-2) - 5$$

$$= -9$$

$$(-2, -9)$$

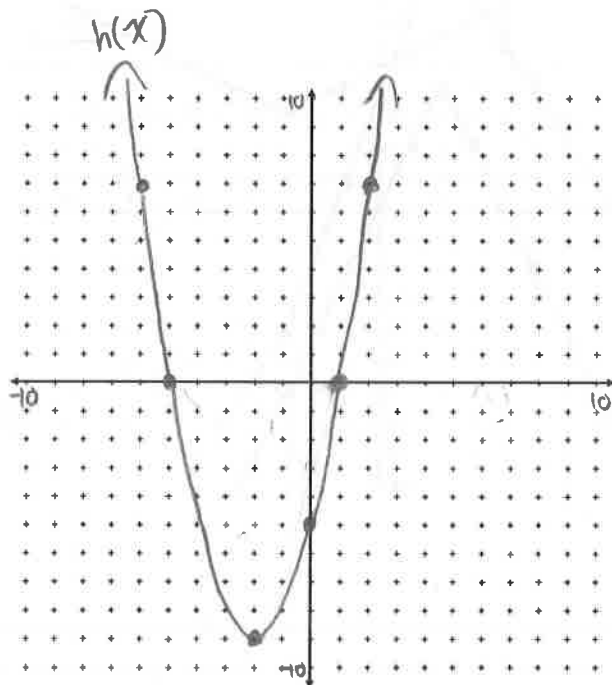
y-int:

$$h(0) = 0^2 + 4(0) - 5$$

$$h(0) = -5$$

$$(0, -5)$$

x	y
-6	7
-5	0
-2	-9
1	0
2	7

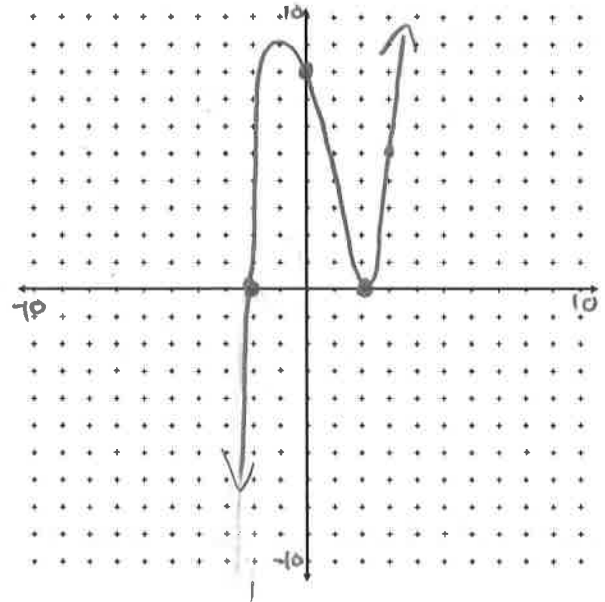


4) Let $f(x) = x - 2$ and $g(x) = x^2 - 4$. Develop an algebraic and graphical model for each of the following:

$$\begin{aligned} \text{a) } y &= f(x)g(x) = (x-2)(x^2-4) \\ &= (x-2)(x-2)(x+2) \\ &= (x-2)^2(x+2) \end{aligned}$$

x -int: $(2,0)$ order 2
 $(-2,0)$ order 1

y -int: $y = (-2-2)^2(-2+2)$
 $y = 8$
 $(0,8)$

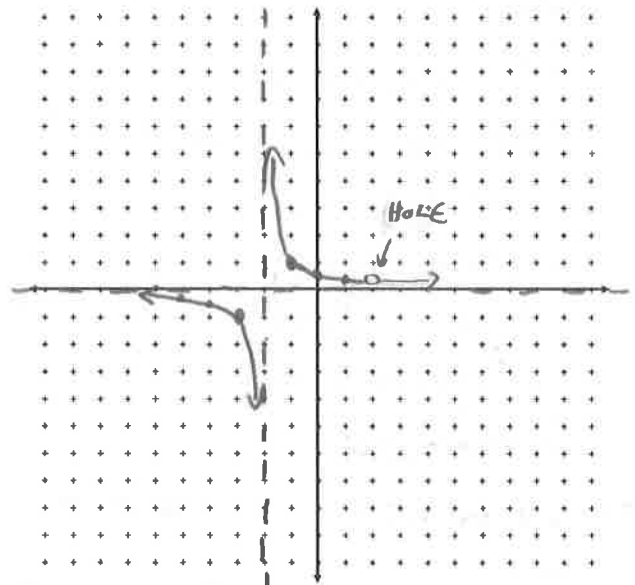


$$\begin{aligned} \text{b) } y &= \frac{f(x)}{g(x)} = \frac{x-2}{x^2-4} \\ &= \frac{x-2}{(x-2)(x+2)} \\ &= \frac{1}{x+2}; x \neq -2, 2 \end{aligned}$$

Hole at $x=2$
 VA: $x=-2$
 HA: $y=0$

x	y
-5	-3
-4	-2
-3	-1
-2	0
-1	1
0	2
1	3

x	$\frac{1}{x+2}$
-5	-0.33
-4	-0.5
-3	-1
-2	und.
-1	1
0	0.5
1	0.33

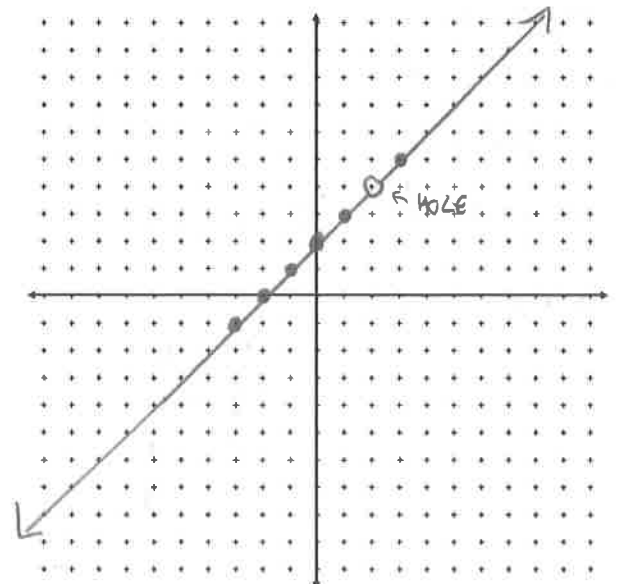


$$\text{c) } y = \frac{g(x)}{f(x)}$$

$$\begin{aligned} &= \frac{x^2-4}{x-2} \\ &= \frac{(x-2)(x+2)}{x-2} \\ &= x+2; x \neq 2 \end{aligned}$$

Hole at $x=2$

x	y
-3	-1
-2	0
-1	1
0	2
1	3
2	undefined
3	5



5) Let $f(x) = x^2 + 2x - 4$ and $g(x) = \frac{1}{x+1}$.

a) Evaluate $g(f(0))$

$$\begin{aligned} f(0) &= 0^2 + 2(0) - 4 \\ f(0) &= -4 \end{aligned} \quad \left\{ \begin{aligned} g(f(0)) &= g(-4) \\ &= \frac{1}{-4+1} \\ &= -\frac{1}{3} \end{aligned} \right.$$

b) Evaluate $f(g(-2))$

$$\begin{aligned} g(-2) &= \frac{1}{-2+1} \\ g(-2) &= -1 \end{aligned} \quad \left\{ \begin{aligned} f(g(-2)) &= f(-1) \\ &= (-1)^2 + 2(-1) - 4 \\ &= -5 \end{aligned} \right.$$

6) Let $f(x) = x^2 + 3x$ and $g(x) = 2x - 5$. Determine an equation for each composite function and graph it.

a) $y = f(g(x))$

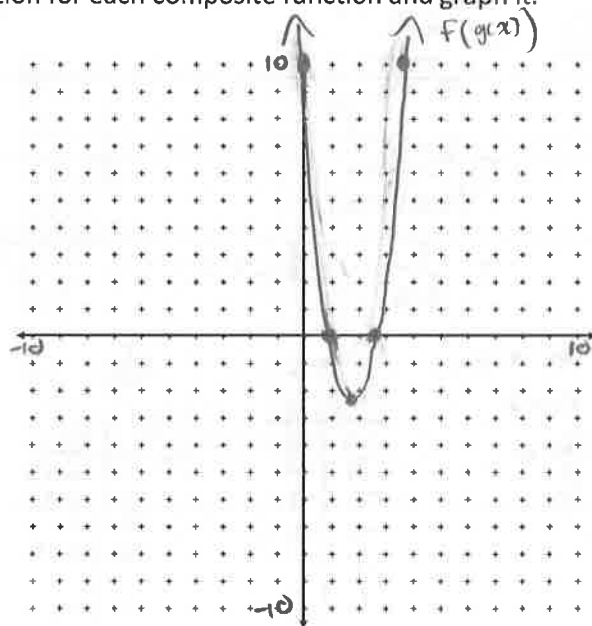
$$\begin{aligned} f(g(x)) &= (2x-5)^2 + 3(2x-5) \\ &= 4x^2 - 20x + 25 + 6x - 15 \\ &= 4x^2 - 14x + 10 \\ &= 2(2x^2 - 7x + 5) \\ &= 2[2x^2 - 5x - 2x + 5] \\ &= 2[x(2x-5) - 1(2x-5)] \\ &= 2(2x-5)(x-1) \end{aligned}$$

x-int: $x = 1$ and $x = 2.5$

x-vertex at $\frac{1+2.5}{2} = 1.75$

y-int: $(0, 10)$

y-vertex $= -2.25$



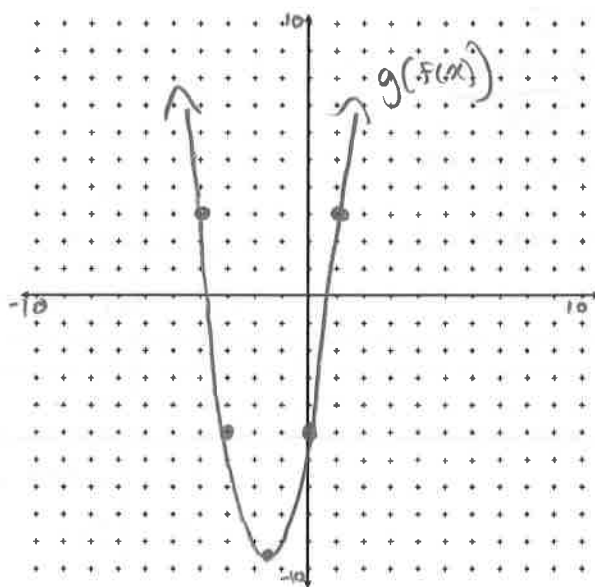
b) $y = g(f(x))$

$$\begin{aligned} g(f(x)) &= 2(x^2 + 3x) - 5 \\ &= 2x^2 + 6x - 5 \end{aligned}$$

x-vertex at $\frac{-b}{2a} = \frac{-6}{2(2)} = -1.5$

y-vertex $= -9.5$

x	y
-4	3
-3	-5
-1.5	-9.5
0	-5
1	3



$$c) y = g(g(x))$$

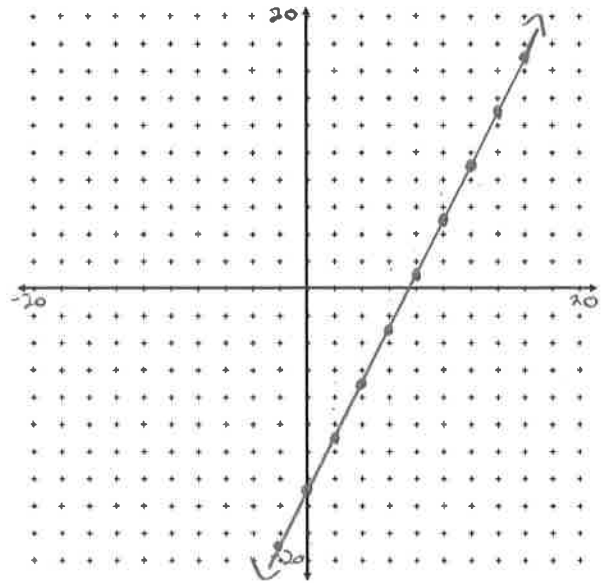
$$= 2(2x - 5) - 5$$

$$= 4x - 10 - 5$$

$$= 4x - 15$$

$$m = 4$$

$$b = -15$$



$$y = g^{-1}(g(x))$$

$$g^{-1}(x) =$$

$$y = 2x - 5$$

$$x = 2y - 5$$

$$\frac{x+5}{2} = y$$

$$g^{-1}(x) = \frac{x+5}{2}$$

$$g^{-1}(g(x)) = \frac{(2x-5)+5}{2}$$

$$= \frac{2x}{2}$$

$$= x$$

$$m = 1$$

$$b = 0$$

